

Identifying and predicting situations of financial stress using heatmaps for Nonbank Financial Institutions in Romania

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Abstract:

Focusing on Nonbank Financial Institutions (NFI), we design a framework for monitoring systemic risk in Romania's financial industry. We start by identifying six risks that NFIs are most likely to face and we construct a set of many indicators that describe, measure, and predict these risks. Using standardized data from the National Bank of Romania (NBR) database, we track the evolution of these indicators over time and derive summary measures. This paper uses Bayesian Model Averaging to select the indicators with the largest prediction power, and the partial Area Under the ROC curve methodology to identify the optimal threshold for every indicator. Contrary to our initial hypotheses, we discover that looking at non-performing loans overall, and their dynamics over time represents a more efficient method of identifying potential vulnerabilities, as opposed to looking into the behavior of different loan categories. At the end of our analysis, we employ three models, which describe the crediting risk and the foreign currency risk for companies and individuals, using a total number of 27 indicators.

The identified indicators, together with their respective thresholds will be further used in the creation of heatmaps. By monitoring a wide variety of indicators, the resulting heatmap can more effectively illustrate systemic vulnerabilities and the determinants linked to them. With a specific focus on nonbank financial institutions, our heatmap shows the evolution of the chosen indicators between December 2010 and June 2025, attributing colors relative to the registered risk level. Because of its visualization advantages, the heatmap can be a convenient and justified addition to the tools that Romanian authorities, such as the NBR and the National Committee for Macroprudential Oversight (NCMO), already use for redacting macroprudential policies.

1. Introduction

1.1 Nonbank Financial Institutions

Nonbank Financial Institutions (NFI) represent an important sector of the financial system, and their number has continuously increased in Romania over the last 25 years. According to the Financial Stability Report of June 2025 from the National Bank of Romania, Nonbank Financial Institutions owned 5.6 percent of the total assets in the Romanian financial system, in the fourth quarter of 2024.

These financial institutions grant credits on a variety of products, offer different services, including loans, leasing, issue guarantees or make financing commitments, outside the traditional banking framework. Nonetheless, we can highlight several differences between banks and NFIs in the way that they offer services to their clients. In particular, NFIs tend to be faster and more flexible in their credit approval process, as they require less documentation from their clients. At the same time, regular banks offer a wider variety of financial products, including savings accounts and larger loans such as mortgages.

It is essential to note that the National Bank of Romania (NBR) defines NFIs different to other countries. A key distinction is that the term does not include pension funds or insurance companies, as they are regulated by the Financial Supervisory Authority (Autoritatea de Supraveghere Financiară). In accordance with the Law No. 93/2009, NFIs are regulated by the competent authorities, in particular the National Bank of Romania.

Academic papers on similar topics published by various central banks highlight the differences among countries in defining the term "nonbank financial institution". As an example, in the United Kingdom the term was mostly used in academic research or macroprudential policies. Previously, the Bank of England used the term "shadow banking" in order to describe the nonbank financial services. More recently, however, they have updated the term to "non-bank financial intermediation (NBFI)". According to Alder's 2023 speech at the Managed Funds Associations' Global Summit, the term of nonbank financial institutions in the United Kingdom includes: investment funds of all types, insurance companies and pension funds, financial vehicles, financial institutions specialized in lending, leasing and factoring companies, as well as other intermediaries such as brokers, dealers, private placement companies, etc. Therefore, other definitions may be broader than the one proposed by the National Bank of Romania, which has a significant impact on the framework for this report.

1.2 Financial Risks Associated with NFIs

The existing literature that reviews the systemic role of nonbank financial institutions covers both advantages and disadvantages. For example, Elliott, Meisenzahl, and Peydró (2024) suggest that the activity of such institutions benefits the financial system, providing valuable services which finance the real economy and diversify credit supply beyond banks. The International Monetary Fund (2023) also argues that "having a broader set of financial intermediaries with different risk profiles, specialized expertise and time horizons fosters efficiency and allows for diversification of risks". Consequently, NFI play a countercyclical role in providing credit while traditional bank lending contracts are in place, functioning as shock absorbers (Elliott, Meisenzahl, and Peydró, 2024).

However, the same papers explain that even though many NFIs provide functions similar to banks, such as maturity transformation and credit intermediation, they are not subject to the same prudential requirements. For this reason, they can contribute to the accumulation of vulnerabilities. Excessive leverage, large liquidity mismatches, and complex interrelated exposures across the financial system can aggravate stress during times of market disruption (Elliott, Meisenzahl, and Peydró, 2024).

Therefore, this report will focus on six risks that significantly impact the Nonbank Financial Institutions. The first is the crediting risk, defined as the possibility of a client not being able to fulfill their payment obligations according to the contract. Considering that the base activity of NFIs is giving out loans, this is one of the most dangerous risks they face. A high number of non-performing loans may lead to severe losses and may affect the NFIs' capacity to continue their activity. Furthermore, we identify the currency risk, which arises due to potential fluctuations in the exchange rate. If NFIs give out loans and have financing sources in

different currencies, fluctuations can negatively affect their income. The third and fourth identified risks are solvency and profitability. In this case, the institution may not have sufficient funds to cover potential losses and uphold their obligations towards creditors and investors. The NFIs are required to remain profitable in order to maintain their core operations and further attract investments. Another issue faced is the contagion risk, identified when difficulties or failures in one institution will spread to its partners, causing a broader systemic impact. Lastly, financing risk affects nonbank financial institutions as well, and this is also called the liquidity risk. In some situations, the number of quick assets may not be enough to cover client withdrawal requests. NFIs do not have access to deposits the way that traditional banks do, which makes them more vulnerable to liquidity risk, as they depend on financing from other sources, such as investors, bonds, loans.

1.3 Indicators and Mapping Heat

As a consequence, we have designed a comprehensive list of quantitative indicators which will aid us in describing, measuring and preventing the systemic risk associated with nonbank financial institutions in Romania. Indicators are quantifiable variables which help define the intensity, evolution or presence of a risk in the chosen financial sector. They can be split into four different categories as follows: descriptive indicators (which measure the current level of risk), leading (anticipate the occurrence of a future risk), early warning (estimate the probability of a financial crisis) and transmission (measure the dissemination of a risk among linked financial institutions). Moreover, indicators must meet certain requirements such as early warning signaling, stability, and intuition (Drehmann and Juselius, 2014).

The identified indicators, together with their respective thresholds will be further used in the creation of heatmaps. These tools are visual representations of vulnerabilities in the financial system, which often use a set of intuitive colors, such as green, yellow, and red, to describe the risk levels over various time periods. As an example, the Bank of Ireland opts for two ways of displaying the heatmap: a point-in-time snapshot (showing the most recent risk status of each indicator) or as a time-series "ribbon" chart (showing how each indicator's value has changed over time). To create a macroprudential risk heatmap, researchers first define thresholds or baseline values for each indicator. This can be done through various methods, such as scientifically deriving a benchmark, using a historical average, or using a value determined previously by another institution such as the European Banking Authority. Guided by the threshold, the other values are attributed colors based on their associated risk level (Ryan, 2017).

Geršl et al. (2021) highlight that central banks and macroprudential authorities often use heatmaps in order to monitor risk build-up, identify potential threats to financial stability using early warning signals and effectively explain the causes of past financial crises. Cevik, Dibooglu and Kutan (2013) also mention that authorities such as central banks may try to tighten macroprudential policies or make the system more resilient when a lot of indicators flash "red". In addition, empirical research indicates that these composite stress indices typically decline prior to and during crises, offering significant insights for policymakers.

These visuals not only support internal monitoring but also make it easier to communicate risk assessments. For instance, a Federal Reserve study notes that heatmaps "may be helpful in communicating financial stability conditions to a broad audience and facilitating the deliberation of countercyclical macroprudential tools" among decision-makers (Aikman et al., 2017). Therefore, heatmaps have become a vital part of the macroprudential toolkit, helping central banks spot and anticipate financial stress by scanning a wide range of indicators and respond quickly to protect financial stability.

1.4 Problem Definition

Several central banks have designed heatmaps in their recent Financial Stability Reports, showing their use in identifying and communicating emerging pockets of stress. As an example, the Irish Central Bank uses heatmaps to flag when key metrics deviate significantly from historical norms associated with stability (Ryan, 2017). Similarly, the framework introduced by Latvijas Banka covers categories such as external and domestic macro-financial conditions, credit risks of borrowers, and banks' liquidity and solvency metrics. Each risk category in this approach comprises several early warning indicators, enabling a comprehensive but concise assessment of changes in those risk areas (Latvijas Banka, 2018).

The majority of published heatmaps are based on system-wide indicators, thus taking a broader scope rather than isolating particular sectors. In practice, this means that risk metrics related to commercial banks, companies, markets, and nonbank financial institutions are often aggregated. For example, the Latvijas Banka heatmap focuses primarily on banking sector vulnerabilities (e.g., credit institution funding liquidity, asset quality, and capital adequacy), along with borrower solvency trends and macroeconomic risk factors, all in one integrated display (Latvijas Banka, 2018).

Even when non-bank financial institutions are considered in these frameworks, they are rarely examined as a standalone category. For instance, Norges Bank’s systemic risk heatmap spans diverse measures of household and corporate credit imbalances, asset price valuations, and overall financial-sector fragilities in Norway, combining them to track the financial cycle’s evolution as a whole (Arbatli and Johansen, 2017). On one hand, this has been a useful approach so far, because by including a broad set of indicators, a heatmap is better at capturing systemic risks and associated vulnerabilities simultaneously. However, a consequence of this approach is that sector-specific risk signals such as risks unique to NFIs, may be overshadowed or diluted when aggregated in the assessment.

In contrast, this report aims to provide a more comprehensive understanding of the indicators of financial stress, by focusing on the systemic role of nonbank financial institutions in Romania. This narrower strategy represents our contribution to the existing literature, as it provides renewed ideas that support and broaden earlier results. Although the domestic NFI segment is relatively small, a potential combination of vulnerabilities across multiple such institutions could amplify the general effect and induce severe stress, amid unfavorable market conditions. As a result, we emphasize the importance of monitoring risks associated with nonbank financial institutions and their potential to transmit or magnify shocks in the broader financial system.

1.5 Explanatory versus Predictive Modeling

The purpose of this research project is both explanatory and predictive. Given that Nonbank Financial Institutions (NFIs) were recently placed under the supervision of the National Bank of Romania, there is a persisting need for new, automated methods that facilitate a deeper understanding of the influence these institutions have on the Romanian financial environment. Therefore, the aim of this report is to present an integrated approach to determine how NFI data can be utilized to identify and predict situations of financial stress.

This entails adopting both a retrospective and a prospective lens. Firstly, we are interested in understanding whether the activity of Nonbank Financial Institutions propagated the financial risks that led to the financial crises included in our database. In doing so, the National Bank of Romania can prevent future systemic instability through close monitoring, the imposition of stricter regulations, or the evaluation of current restrictions. Additionally, we are interested in associations, specifically whether a shift in particular indicators represents a signal for an approaching crisis.

Using approaches corresponding to explanatory modeling, we conduct a rigorous selection of potential variables based on economic theory and expert-based judgment. Simultaneously, the exploratory data analysis is broad and computationally driven, mirroring a predictive approach. More specifically, we search for any indicator or feature that possesses informational value, in an attempt to expand our initial theoretical construct.

Ultimately, by synthesizing economic intuition with data-driven discovery, this study seeks to provide a robust analytical framework for the National Bank of Romania. This dual approach ensures that the resulting model is not only statistically rigorous but also economically interpretable, equipping supervisors with the necessary tools to proactively mitigate systemic risks arising from Nonbank Financial Institutions.

2. Data Description and Methods

2.1 Indicators

In order to generate a time series set of potential indicators of financial risk, we use a large amount of data, which is available in the NBR database. More specifically, we use the Central Credit Register (CCR), which is a center of data collection within the National Bank of Romania. It operates as a credit bureau, where banks and other financial institutions, such as NFIs, report banking risk data and information on card fraud. All of Romania's credit institutions report information to the CCR and have access to its database.

As mentioned previously, two main categories of information are reported to the CCR. Firstly, we have the Banking Risk Data, which includes details about the loans and other commitments a client (whether an individual or a company) has with a bank. This covers information like the loan amount, repayment term, and any accumulated arrears. Then we have the Card Fraud Information, which concerns fraud committed by both credit and debit cardholders.

As of January 1, 2010, non-bank financial institutions also have access to the CCR, provided they are registered in a Special Register maintained by the National Bank of Romania. Generally, only significant institutions are listed in this register. To qualify, they must meet the following criteria: their equity and borrowed funds must exceed 50 million RON and the value of the loans they have granted must be at least 25 million RON.

Additionally, we use data collected by the Credit Bureau, which is a private company, founded in 2004, with a single purpose: to collect and centralize data on the payment behavior of individuals who take out loans. This information is then made available to financial institutions so they can accurately assess the risk when evaluating a loan application. Almost all commercial banks, authorized non-bank financial institutions, and other institutions report information about their clients on a monthly basis, including: the date a loan was granted, the borrowed amount, the remaining balance, if there have been any payment delays and how significant they were, if any early repayments were made and more.

Therefore, this represents a suitable place to start our analysis. All this data can be further processed in order to obtain indicators which describe and predict the crediting risk and the foreign currency risk, specifically for non-bank financial institutions. In order to do this, we have decided on a "larger to smaller" approach, which entails first looking at 11 indicators for the loans overall, and then further splitting the types of loans into progressively smaller categories, thus making it easier to identify specific vulnerabilities in the system. We are able to calculate the indicators on a monthly basis, from January 2007 until June 2025.

Therefore, we first see that loans offered by non-bank financial institutions can be split into two categories: for companies and for individuals. Companies can be further split into Small and Medium Enterprises (SME) and Corporations. Their loans can also be split by RON and Foreign Currency, Maturity smaller or larger than a year, and many more. Therefore, we start from the 11 indicators described below, and we recalculate them using the same formula, but different values, every time we do another split.

Variable	Description	Formula
y	The binary dependent variable, taking values: 0 if in normal conditions and 1 one year before entering a crisis period	
Date	Date format MM/DD/YYYY, where DD represents the last day of the month, MM the month, and YYYY the year for which the value of each indicator was calculated	
Crediting Risk		
v1	NPL Ratio (%)	No. of NPL credits / Total stock $\times 100$

Variable	Description	Formula
v2	Share of new credit (%)	New credit stock / Total stock $\times$ 100
v3	Share of new debtors (%)	No. of new debtors / Total No. of debtors $\times$ 100
v4	Credit stock dynamics (%)	(Total credit stock in year X - Total credit stock in year Y-1) $\times$ 100 / Total credit stock in year Y-1, where Year 2007 = 0.
v5	Non-performing credit stock dynamics (%)	(Total NPL credit stock in year X - Total NPL credit stock in year Y-1) $\times$ 100 / Total NPL credit stock in year Y-1, where Year 2007 = 0.
v6	Debtors dynamics (%)	(No. of debtors in month X - No. of debtors in month Y-1) $\times$ 100 / Total No. of debtors in month Y-1, where Year 2007 = 0.
v7	Non-performing debtors dynamics (%)	(No. of non-performing debtors in month X - No. of non-performing debtors in month Y-1) $\times$ 100 / Total No. of non-performing debtors in month Y-1, where Year 2007 = 0.
v8	New credit per debtor	New credit stock / No. of new debtors
v9	Share of non-performing debtors (%)	No. of non-performing debtors / Total No. of debtors $\times$ 100
Foreign Currency Risk		
v10	Share of foreign currency credit (%)	Total foreign exchange credit stock / Total stock $\times$ 100
v11	Share of debtors with foreign currency credit (%)	Total No. of debtors with foreign exchange credit / Total No. of debtors $\times$ 100

Table 1: The 11 main indicators for financial risks

2.2 Bayesian Model Averaging

In a crisis prediction setting with many candidate indicators, there is significant uncertainty about which regressors matter. Choosing a single model based on expert judgment risks omitting important indicators or including irrelevant ones. For this reason, we work with Bayesian Model Averaging (BMA), which will determine the predictive capacity of every indicator. This framework exhaustively simulates all combinations of the explanatory variables and averages them according to their posterior probabilities. This approach directly addresses model uncertainty by incorporating the information from a vast model space rather than relying on one manually chosen set of regressors. As a result, BMA minimizes subjective judgment in model selection and automatically detects the subset of the most important predictors from the data (Babecký et al., 2013).

The selected target variable is a binary crisis indicator that marks historical crisis episodes on a monthly timeline, and it is constructed using the Country-Level Index of Financial Stress (CLIFS). The CLIFS includes six, mainly market-based, financial stress measures that capture three financial market segments: equity markets, bond markets and foreign exchange markets. In addition, when aggregating the sub-indices, the CLIFS takes the co-movement across market segments into account (Duprey, Klaus, and Peltonen, 2017). One key application of CLIFS is to define and calibrate thresholds that distinguish normal conditions from crisis ones. Because it is a continuous index, analysts often need to calibrate it into binary signals, which will also be done in this paper. Therefore, in addition to the time-series data we have regarding the values of the indicators, we also have a list of binary variables which describe whether a crisis was approaching or occurring at a specific point in time.

Some theoretical aspects of implementing the BMA framework include:

- A pool of equations for every dependent variable
- K independent variables
- A self defined limit of L predictors
- The total number of combinations of the exogenous variables which compile the model space of BMA: $\sum_{l=1}^L \frac{K!}{l!(K-l)!}$

The key output of the Bayesian Model Averaging framework is the Posterior Inclusion Probability (PIP) of each indicator. Calculating the posterior probability involves using the Bayes Rule:

$$g(\beta | y) = \frac{g(\beta) f(y | \beta)}{f(y)}$$

where:

- $g(\beta | y)$ is the posterior probability
- $f(y)$ is the function prior to the data
- $f(y | \beta)$ is the likelihood function
- $g(\beta)$ is the initial function of the parameter

We can interpret PIPs as a guide to the most efficient indicators. What Bayesian Model Averaging suggests is that an indicator with a high PIP is statistically significant in a high fraction of models, making it a reliable predictor of the crisis variable. Classifying a posterior probability as high is based on expert judgment. For example, Vidal-Abarca and Ruiz (2015) treat an indicator with $PIP > 0.7$ as an effective predictor.

Other researchers have also used Bayesian Model Averaging in order to identify which risk indicators are the most informative in signaling financial crises. For instance, Babecký et al. (2013) obtained that only about one-third of the 30 candidate variables were robust predictors of crises. This highlights how BMA sifts out the noise variables, effectively reducing bias. Overall, the findings derived from high posterior probability values guide policymakers on which vulnerabilities to monitor.

2.3 Optimal Thresholds

In order to identify the optimal threshold for every indicator, we will use two key metrics: the Receiver Operating Characteristic (ROC) curve and its associated Area Under the ROC Curve (AUROC). The ROC curve is an analytical method, represented as a graph, used to evaluate the performance of a binary, crisis identification method (Nahm, 2022). Similarly to what we did for Bayesian Model Averaging, we set the target variable as a binary crisis indicator that marks historical crisis episodes on a monthly timeline.

The ROC and AUROC are based on Type I and Type II errors, which, in statistical hypothesis testing, represent two distinct ways that a conclusion can be wrong. In the case of financial risk indicators, a Type I error occurs when the indicator gives off a warning signal, even though no crisis situation occurred. Conversely, a Type II error occurs when the indicator fails to detect a crisis situation, and does not give off

a warning signal when it should. Using the confusion matrix, which is described below, we can calculate the true positive rate and the false positive rate:

	Signal	No Signal
Crisis	A	B
No Crisis	C	D

Concept	Definition	Formula
Sensitivity	The proportion of actual positives correctly identified by the test	$\frac{A}{A+B}$
Specificity	The proportion of actual negatives correctly identified by the test	$\frac{D}{C+D}$
False Negative	The proportion of actual positives that the test incorrectly classifies as negative	$\frac{B}{A+B}$
False Positive	The proportion of actual negatives that the test incorrectly classifies as positive	$\frac{C}{C+D}$
Accuracy	The overall proportion of correct predictions (both positives and negatives)	$\frac{A+D}{A+B+C+D}$

Table 2: Key formulas derived from the Confusion Matrix

The ROC curve is drawn using $1 - \text{specificity}$ on the x-axis and sensitivity on the y-axis. Interestingly, the paper published by Nahm in 2022 states that: "The ROC curve was initially developed to determine between a signal (true positive result) and noise (false positive result) when analyzing signals on a radar screen during World War II.". Since then, statisticians have developed multiple uses for the ROC curve, in medicine, psychology, bio-informatics, machine learning and finance. In our case, it is used to assess the overall warning performance of an indicator and select an optimal threshold for determining the presence or absence of a financial crisis.

Given the inverse relationship between sensitivity and specificity, policymakers notice that applying looser criteria to increase one of them results in the decrease of the other. Therefore, it is essential to set the optimal threshold for every indicator, by finding the appropriate compromise between the two (Nahm, 2022). For this reason, we will also use the Partial AUROC, which is the area under the ROC curve calculated over a restricted range, based on the policymakers' preferences.

As previously mentioned, identifying an optimal threshold for crisis prediction involves deciding which balance between true positives and false positives is desired. One way to do this is by optimizing the policy makers' loss function, defined by Alessi and Detken (2014) as follows: $L(\theta, \tau) = \theta * T_1(\tau) + (1 - \theta) * T_2(\tau)$. We call θ the preference weight, $0 \leq \theta \leq 1$, $T_1(\tau)$ the true positive rate and $T_2(\tau)$ the false positive rate associated with the threshold τ .

The optimal threshold in this framework is the one that minimizes $L(\theta)$, and will depend on the chosen θ . The same procedure is used by Lang et al. (2018): for each candidate threshold τ , they compute $T_1(\tau)$ and $T_2(\tau)$ using historical data. In the end, they pick the τ^* that yields the lowest $L(\theta, \tau)$.

The function's definition implies that a higher θ places more weight on avoiding missed crises. Varying θ traces out a family of optimal trade-off points on the ROC curve. The partial AUROC analysis ensures the chosen threshold is not based on extreme, implausible preferences, by constraining θ to a reasonable range. Moreover, any indicator with a pAUROC value lower than 0.5 is considered uninformative. On the other hand, higher pAUROC values signal that an indicator consistently provides useful warnings within the acceptable false-alarm range.

2.4 Heatmap

To integrate our analytical results, we will use a heatmap, constructed as a two-dimensional color-coded matrix that visualizes the risk level registered by every indicator throughout a period of several years. The heatmap provides an intuitive summary of how indicators behaved before, during and after past financial crises (Salvati et al., 2023).

Furthermore, after running the Bayesian Model Averaging on all candidate variables, we will obtain a list of Posterior Inclusion Probabilities, and we can identify the subset with the highest PIPs. The indicators which ranked high in our analysis will become the prime candidates for early warning signals, because the framework evaluated them as most likely to be selected in good predictive models. For visualization purposes, we treat each of the selected indicators as a row in the heatmap.

As explained above, we will determine an optimal risk threshold for every indicator using the partial Area Under the ROC Curve (pAUROC) method. The derived cut-point represents the boundary for what we consider “low risk” and “high risk” values for each indicator. Each indicator’s threshold will be estimated separately, reflecting its unique relationship with the outcome.

For the risk categorization and color encoding, we use a three color scheme. This intuitively highlights the severity of the risk: blue denotes a safe or negligible risk level, red denotes a significant risk, and beige lies in between.

Since we are summarizing individual indicators’ risk levels, the columns of the heatmap represent the years for which we have calculated their corresponding values. As a result, every cell represents the intersection of an indicator and a year, and gets colored blue, beige, or red, based on the calculated risk level.

The final heatmap thus condenses complex statistical results into a single image: a grid of colored cells, where color intensity denotes risk magnitude. This visualization method leverages the strength of heatmaps in revealing patterns and outliers in multivariate data. By scanning the heatmap, the audience can immediately identify the indicators which described the largest risk and those with minimal impact, as well as the corresponding point in time (Salvati et al., 2023).

3. Results

3.1 Indicators

As mentioned in the above section, we started our analysis by looking at the loans that non-bank financial institutions can offer. We are interested in four subcategories, based on the debtors: companies in total, small and medium enterprises (SMEs), corporations, and individuals.

Then, for companies in total, SMEs and corporations, we see that the loans they take out can be split by different criteria, which are described below:

- By currency: RON and Foreign Currency
- By maturity: maturity < 365 days or maturity >= 365 days
- By activity sector: agricultural sector, extractive industry, manufacturing industry, utilities, construction, trade, services and real estate
- By technical activity: low tech, medium low tech, medium high tech, high tech, less-knowledge intensive services and knowledge intensive services

We are interested in how each of these subcategories behave in relation to the crediting risk. Because they all represent a perfect split of the total share of loans (meaning that when summed up, they equal the same total number), an additional step was required in order to avoid multicollinearity or high correlations in our statistical models. More specifically, we only assess the loans with: foreign currency, maturity >= 365 days, agricultural sector, extractive industry, manufacturing industry, utilities, construction, trade, services, low tech, medium low tech, medium high tech, high tech, and less-knowledge intensive services.

Consequently, we have 14 types of loans, all independent in order to avoid multicollinearity and for each, we calculate the 9 indicators corresponding to crediting risk, obtaining a total set of 126 indicators. This process is repeated for companies as a total, SMEs and corporations.

In the case of loans given out for individuals, we focus on fewer categories, assessing the loans with foreign currency and the ones for the real estate sector. Now we have two types of loans, independent in order to avoid multicollinearity and for each, we calculate the 9 indicators corresponding to crediting risk. We obtain a total set of 18 indicators, which is significantly narrower, compared to the set for companies.

Indicator	mean	median	min	max
NPL Ratio (%)	9.10	6.17	0.00	20.89
Share of new credit (%)	3.23	3.46	0.00	14.95
Share of new debtors (%)	4.54	4.93	0.00	9.76
Credit stock dynamics (%)	1755.47	12.18	-19.42	41094.67
Non-performing credit stock dynamics (%)	9787.01	-3.17	-32.51	1372477.23
Debtors dynamics (%)	4567.32	11.11	-26.07	104530.49
Non-performing debtors dynamics (%)	14218.97	-5.43	-20.61	1052300.00
New credit per debtor	227084.09	213760.79	0.00	1006440.00
Share of non-performing debtors (%)	8.21	5.67	0.00	18.29
Share of foreign currency credit (%)	87.46	86.44	56.68	99.60
Share of debtors with foreign currency credit (%)	87.69	85.74	56.25	99.93

Figure 1: Summary statistics of the main 11 indicators for companies in total

Indicator	mean	median	min	max
NPL Ratio (%)	9.82	6.24	0.00	22.32
Share of new credit (%)	3.17	3.38	0.00	15.01
Share of new debtors (%)	4.47	4.83	0.00	9.76
Credit stock dynamics (%)	1440.07	12.83	-19.88	33298.44
Non-performing credit stock dynamics (%)	9241.10	-3.36	-34.54	1291798.84
Debtors dynamics (%)	4541.49	10.82	-26.89	103896.30
Non-performing debtors dynamics (%)	13829.96	-5.39	-21.22	1023500.00
New credit per debtor	201508.44	189355.24	0.00	1006440.00
Share of non-performing debtors (%)	7.85	5.33	0.00	17.32
Share of foreign currency credit (%)	6.22	4.49	0.00	22.52
Share of debtors with foreign currency credit (%)	4.41	4.68	0.00	13.94

Figure 2: Summary statistics of the main 11 indicators for SMEs

Indicator	mean	median	min	max
NPL Ratio (%)	4.79	2.37	0.00	16.97
Share of new credit (%)	3.08	3.35	0.00	7.40
Share of new debtors (%)	7.40	8.51	0.00	13.71
Credit stock dynamics (%)	93066.48	4.64	-21.39	2326381.50
Non-performing credit stock dynamics (%)	38.82	0.00	-66.40	6900.93
Debtors dynamics (%)	4826.35	0.22	-15.47	109700.00
Non-performing debtors dynamics (%)	23.39	-5.54	-30.00	1022.22
New credit per debtor	918492.71	968132.62	0.00	2578515.42
Share of non-performing debtors (%)	19.39	15.87	0.00	43.13
Share of foreign currency credit (%)	85.82	90.21	0.00	100.00
Share of debtors with foreign currency credit (%)	86.28	91.96	0.00	100.00

Figure 3: Summary statistics of the main 11 indicators for corporations

Indicator	mean	median	min	max
NPL Ratio (%)	6.77	7.47	0.00	15.91
Share of new credit (%)	4.21	4.21	0.00	21.82
Share of new debtors (%)	3.72	3.39	0.00	16.08
Credit stock dynamics (%)	44.88	9.66	-65.95	581.55
Non-performing credit stock dynamics (%)	376.50	9.83	-33.38	20462.68
Debtors dynamics (%)	321.37	0.58	-57.82	12660.35
Non-performing debtors dynamics (%)	10411.89	3.83	-42.73	730158.33
New credit per debtor	34451.55	9684.28	0.00	255350.27
Share of non-performing debtors (%)	16.57	20.04	0.00	29.55
Share of foreign currency credit (%)	36.21	20.38	0.00	99.97
Share of debtors with foreign currency credit (%)	24.05	1.77	0.00	100.89

Figure 4: Summary statistics of the main 11 indicators for individuals

3.2 Bayesian Model Averaging

In the next step of our analysis, we applied the BMA to the eight models described above, which allowed us to identify which indicators are the most sustainable. We obtained the posterior inclusion probabilities as an output of this algorithm, and we are interested in the coefficients of the uniform prior, because we attribute equal weights to any random scenario generated by the BMA.

Additionally, we require the PIP to be larger than 0.7 in the case of "total" models, and equal to 1 in the case of "split" models. This is because in a model with only 11 indicators, at most one has a PIP equal to 1, and the differences between PIPs grow significantly larger as they decrease. Therefore, we consider an indicator to have a satisfactory prediction power if it can successfully signal at least 70 percent of the randomly generated crises. Conversely, in a model with 126 indicators, where the set of possible combinations is much larger, we saw that more indicators had a PIP value of 1. Moreover, even as these values decrease, the differences between them are not perceptible. In the case of "split" models, we have approximately 20 indicators with PIPs equal to 1, and they gradually lower to 0.99, 0.98 and so forth.

Consequently, we identify the following list of significant indicators, which have registered a high prediction power in our analysis of "total" models:

- Companies in Total: Share of foreign currency credit, share of debtors with foreign currency credit, NPL Rate
- SMEs in Total: Share of foreign currency credit, share of debtors with foreign currency credit, Share of non-performing debtors
- Corporations in Total: Share of foreign currency credit, share of debtors with foreign currency credit, Non-performing credit stock dynamics, Non-performing debtors dynamics
- Individuals in Total: Non-performing debtors dynamics

Given these results, we can draw the following conclusions: in all three models which analyzed the crediting behavior of companies, we see that loans taken in foreign currency constitute the largest risk predictors. In almost all simulations, indicators such as share of foreign currency credit and share of debtors with foreign currency credit managed to accurately describe and predict financial crises. Therefore, they should be studied in more detail and we decide to expand our list of such indicators. Instead of using the two proposed initially, we calculate the 9 crediting risk indicators only for loans taken in a foreign currency and thus create a model for Companies in Foreign Currency for the following steps of our analysis. This approach allows us to properly cover all vulnerabilities which might arise for non-bank financial institutions which give out loans in one currency, while building their capital on another.

Another important result is identifying the advantage of using a "total" model in the case of companies, instead of a "split" one. We reach this conclusion because the indicators with the highest PIPs in the case of "split" models were almost always the following four: credit stock dynamics, debtor dynamics, non-

performing credit stock dynamics and non-performing debtors dynamics, selected continuously for most loan categories. The reason for splitting was to identify whether or not a certain loan category could generate more vulnerabilities than another. This is the case for loans in foreign currency as opposed to loans in RON, but for all other categories, this difference isn't revealed anymore. Instead, we conclude that most vulnerabilities are generated by non-performing loans, by how their number changes over time, as well as how the influx of debtors evolves. Consequently, we employ another model for the next stage: Companies in Total, which is described by the 9 crediting risk indicators calculated at first.

Lastly, in the case of individuals, both the "total" and the "split" models show few indicators as having a high prediction power. Because this perspective is also important for the final heatmap, we will further employ a Individuals in Total model as well. Since the loans given out by non-bank financial institutions are split between individuals and companies, and we already plan to create a framework for companies, it is significant to visualize the situation for individuals as well.

3.3 Optimal Thresholds

Since our data could only be calculated for a limited period in time, namely from January 2007 until June 2025, we could only capture three crises: the global financial crisis from 2008-2009, European Union sovereign debt crisis from 2011-2012, and the COVID pandemic in 2019-2020. For this reason, we acknowledge that the results of the partial AUROC analysis might be slightly deviated, because of the lack of reference crises. The National Bank of Romania had previously established thresholds for the following crediting risk indicators: NPL ratio, credit stock dynamics, and nonperforming credit stock dynamics. Therefore, we first created the following dashboard, using the European Banking Authority (EBA) framework. Here we can see what the value was for each indicator on December 31st, yearly, from 2017 until 2024. The last column of the table represents the value registered for June 2025.

For companies in total, we see that the NPL ratio mostly lies above the 3 percent signaling threshold. It steps down, however, from 6.6 percent in 2017 to lower values by 2022–2024, then ticks back up to 3.1 percent in June 2025. Credit stock dynamics shows two distinct phases: subdued growth through 2021, followed by an acceleration in 2022–2024 when the indicator stays above 15 percent and peaks at 19.4 percent in 2023. By mid-2025 growth cools to 12.1 percent, but still remains positive. More worrying is the non-performing credit stock: after steady improvements in 2017–2022 (mostly double-digit declines), it turns positive in 2023 (10.9 percent) and then jumps into the high-risk zone in December 2024 to June 2025 (27.5 and 27.1 percent), indicating a clear deterioration in loan quality.

A similar pattern holds for companies taking loans in foreign currency, but with a stronger quality signal in the headline NPL. The NPL ratio falls below three percent from 2022 onward (between 2.5 and 2.7 percent through June 2025), pointing to a generally good performance. At the same time, credit stock dynamics are volatile: mostly under 15 percent, but surging to 23.9 percent in 2023 and remaining elevated at 15.9 percent in 2024 before easing to 12.1 percent in 2025. The non-performing credit stock dynamics confirms rising risk late in the period: after sustained improvements through 2022, they flip to 5.9 percent in 2023 (warning zone) and then escalate sharply to 25.6 percent in 2024 and 31.6 percent by June 2025, lying firmly in the systemic-risk category.

The table showing loans given out to individuals is structurally weaker. The NPL ratio hovers near the top of the 3 to 8 percent band for most years and breaches 8 percent in 2020 and again in June 2025, underscoring persistent household repayment pressure. Credit stock dynamics are cyclical: contractions in 2018 and 2022 are followed by periods of growth, namely 2019 (16.1 percent), 2020 (15.1 percent), 2024 (16.7 percent) and 2025 (18.1 percent), placing the indicator repeatedly in the early-warning zone. The dynamics of non-performing credit stock changes widely: severe increases in 2018–2020 (20.1 32.5 percent), a brief improvement in 2021–2022 (down to 3.3 and 3.9 percent), and then renewed stress in 2023–June 2025 (13.4, 11.3 and 20.7 percent), keeping the metric in high-risk territory in the latest readings.

Risk indicators	Threshold	2017	2018	2019	2020	2021	2022	2023	2024	June 2025
Crediting Risk										
NPL Ratio	< 3%									
	[3% – 8%]	6.6	5.3	4.3	3.7	3.4	2.6	2.4	2.7	3.1
	> 8%									
Credit stock dynamics	< 15%									
	[15% – 20%]	13.1	14.7	11.6	2.7	7.1	16.6	19.4	15.1	12.1
	> 20%									
Non-performing credit stock dynamics	< 5%									
	[5% – 10%]	-27.7	-7.7	-8.1	-12.5	-0.5	-11.1	10.9	27.5	27.1
	> 10%									

Table 3: EBA Dashboard for Companies in Total

Risk indicators	Threshold	2017	2018	2019	2020	2021	2022	2023	2024	June 2025
Crediting Risk										
NPL Ratio	< 3%									
	[3% – 8%]	6.9	5.7	4.4	3.7	3.4	2.5	2.1	2.3	2.7
	> 8%									
Credit stock dynamics	< 15%									
	[15% – 20%]	10.1	13.7	13.2	3.9	7.2	14.3	23.9	15.9	12.1
	> 20%									
Non-performing credit stock dynamics	< 5%									
	[5% – 10%]	-33.1	-5.1	-12.1	-13.9	-0.6	-15.4	5.9	25.6	31.6
	> 10%									

Table 4: EBA Dashboard for Companies in Foreign Currency

Risk indicators	Threshold	2017	2018	2019	2020	2021	2022	2023	2024	June 2025
Crediting Risk										
NPL Ratio	< 3%									
	[3% – 8%]	5.1	6.8	7.8	8.1	7.3	7.7	7.8	7.4	8.1
	> 8%									
Credit stock dynamics	< 15%									
	[15% – 20%]	14.4	-9.1	16.1	15.1	7.3	-1.5	12.9	16.7	18.1
	> 20%									
Non-performing credit stock dynamics	< 5%									
	[5% – 10%]	-0.9	20.1	32.5	20.2	-3.3	3.9	13.4	11.3	20.7
	> 10%									

Table 5: EBA Dashboard for Individuals in Total

We can, furthermore, analyze the partial AUROC results, which provide important insights in the case of all three models. Identified optimal cut-off points highlight the thresholds beyond which systemic vulnerabilities are likely to materialize, thereby offering a quantitative basis for the design of early warning systems.

In the case of NPL ratios, we consider any value above 3 percent as a potential warning signal. These results are consistent with the previous framework employed by the National Bank of Romania. The same conclusion can be reached for the credit stock dynamics. However, our analysis does not align with the previous framework in the case of non-performing credit stock dynamics. This results from the fact that, as stated above, our data for crises is limited, and it might be more useful to consider an expert-based threshold for this indicator in particular.

When looking at the other indicators, we notice that the share of non-performing debtors has a threshold of 1 percent across all three models. Therefore, it proves to be highly informative and it confirms that non-bank financial institutions are particularly sensitive to borrower quality and repayment behavior. Finally, the role of debtor dynamics and the growth rate of credit per debtor, although marked by high numerical thresholds, point to the systemic risks associated with rapid expansion in the client base and concentrated increases in average credit exposures.

Variable Name	Optimal Cutoff
NPL Ratio	2
Share of new credit	5
Share of new debtors	6
Credit stock dynamics	22
Non-performing credit stock dynamics	-23
Debtor dynamics	100867
Non-performing debtors dynamics	62
Growth rate of credit per debtor	334676
Share of non-performing debtors	1

Table 6: pAUROC Results for Companies in Total

Variable Name	Optimal Cutoff
NPL Ratio	2
Share of new credit	4
Share of new debtors	6
Credit stock dynamics	25
Non-performing credit stock dynamics	-24
Debtor dynamics	89
Non-performing debtors dynamics	72
Growth rate of credit per debtor	342182
Share of non-performing debtors	1

Table 7: pAUROC Results for Companies in Foreign Currency

Variable Name	Optimal Cutoff
NPL Ratio	3
Share of new credit	7
Share of new debtors	8
Credit stock dynamics	20
Non-performing credit stock dynamics	10
Debtor dynamics	-45
Non-performing debtors dynamics	60
Growth rate of credit per debtor	38227
Share of non-performing debtors	1

Table 8: pAUROC Results for Individuals in Total

3.4 Heatmap

The generated heatmaps showcase the evolution of the 9 crediting risk indicators from December 2010 until June 2025. We chose this shorter time frame, even though it does not capture the financial crisis of 2008-2009, because of Regulation no. 20 of 2009, which was issued by the National Bank of Romania (BNR) at the end of 2009. This is a crucial piece of legislation that clarifies and implements Law no. 93/2009 regarding non-banking financial institutions (NFI). It sets strict requirements for capital adequacy, liquidity, and risk management and thus, it is designed to ensure financial stability and protect customers. It also clarifies reporting obligations, as it mandates that NFIs report any significant changes to their status to the BNR (e.g., changes in ownership, management, or business activities).

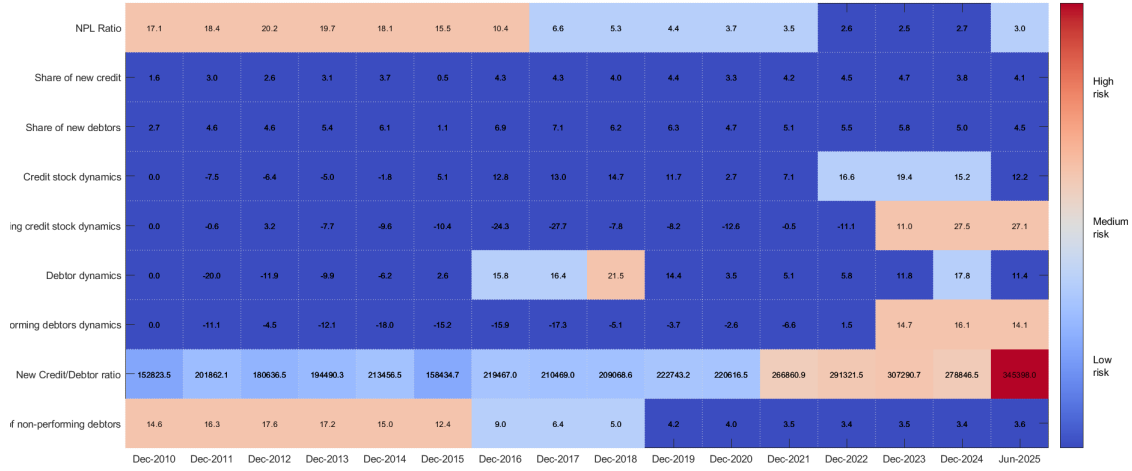


Figure 5: Heatmap for Companies in Total

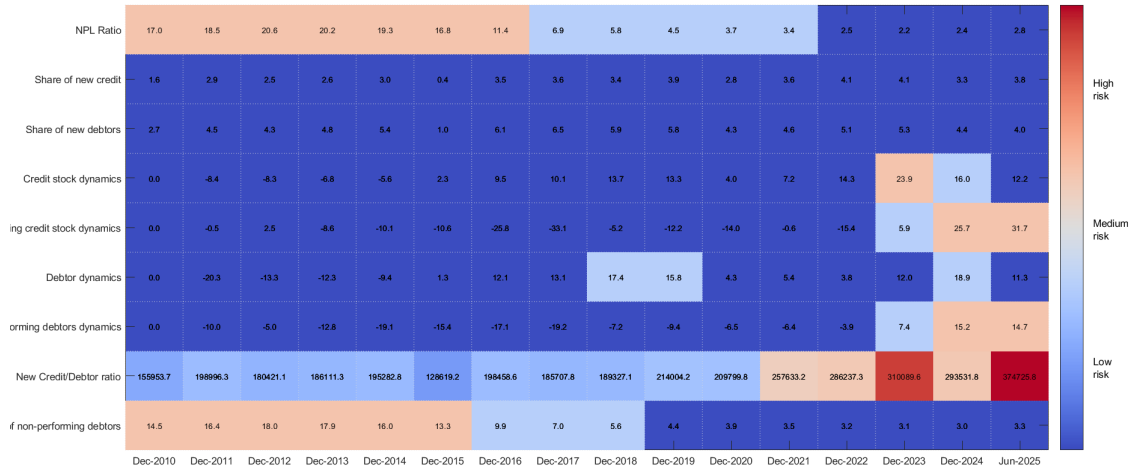


Figure 6: Heatmap for Companies in Foreign Currency

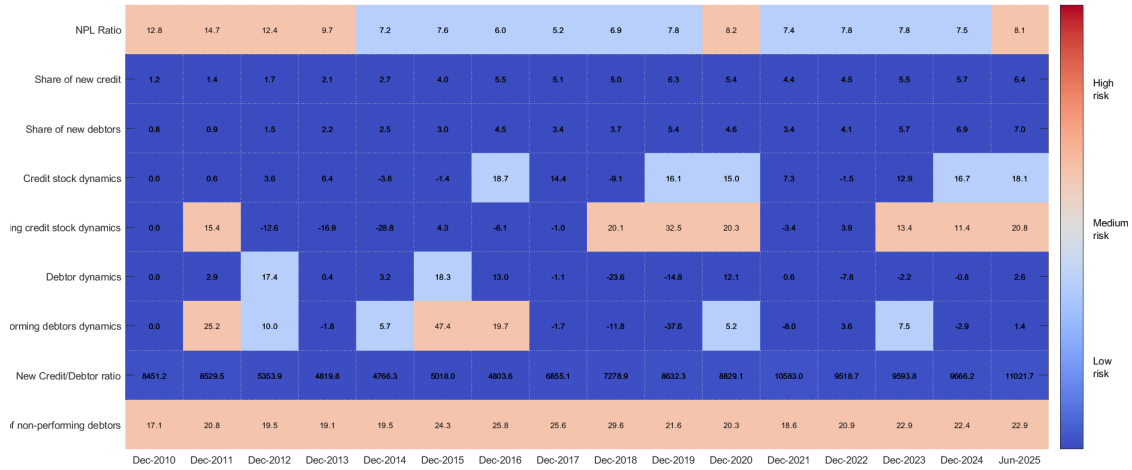


Figure 7: Heatmap for Individuals in Total

Figures 5 to 7 present the heatmaps of the crediting risk indicators for loans extended by nonbank financial institutions to companies in total, companies in foreign currency, and individuals, respectively, between 2010 and June 2025. For each year, we report December values, with the series closing mid-2025. As before, the color scheme is intuitive: blue denotes a safe or negligible risk level, red indicates significant risk, and intermediate shades capture the transition from medium to higher vulnerability. Looking across the three panels, a clear differentiation emerges between the exposures of companies and those of individuals. Figure 5 highlights that companies in total have seen repeated episodes of stress, with the NPL ratio firmly in the high-risk zone up until 2016 before a sustained improvement. At the same time, the later years point again to vulnerabilities: credit stock dynamics accelerated in 2022–2023, while non-performing credit growth spiked sharply in 2024 and mid-2025, situating risk back in the systemic category.

For companies borrowing in foreign currency, the heatmap reveals a broadly similar pattern but with an even stronger deterioration in recent years. The NPL ratio shows steady improvement, dropping into the safe

zone after 2021, but this masks mounting vulnerabilities elsewhere. Credit and non-performing credit stock dynamics deteriorated substantially in 2023–2025, with the latter climbing decisively into high-risk territory. These results underline the destabilizing role of currency mismatches, where borrowers face heightened repayment pressures during episodes of exchange rate volatility. On the contrary, the profile of individuals is more moderate, but still reveals cyclical strains. The NPL ratio remained elevated, often near or above the medium-risk threshold, while credit growth for households displayed recurrent bursts in 2017, 2019, 2024, and 2025. Non-performing credit dynamics also show repeated spikes, most notably in 2018–2020 and again from 2023 onwards, even though the magnitude remains less synchronized and somewhat less severe than in the corporate sector.

Consequently, the heatmap analysis provides valuable insights into the build-up of risk across borrower segments and lending structures, offering a visual early warning tool for financial stress. These results highlight the importance of strengthening the monitoring of nonbank financial institutions, given their growing role in credit provision, and implementing targeted macroprudential measures to mitigate systemic risk.

4. Discussion and Conclusion

The result of our analysis is developing a heatmap-based monitoring framework for Romanian nonbank financial institutions using 9 indicators across the two most significant risk categories. The framework employs partial AUROC thresholds and Bayesian Model Averaging to flag early warnings. We find that credit-quality and currency-mismatch indicators are the most predictive of NFI stress.

Several recent studies have used similar heatmap or dashboard approaches to systemic risk, reinforcing our findings. For instance, Venditti et al. (2018) construct an Italian risk dashboard with about 70 indicators grouped into seven risk categories. They likewise monitor crediting indicators, including NPL rates for companies and individuals, and find them to have increased prediction power in highlighting risk build-up. Similarly, Geršl et al. (2021) find that macroeconomic and foreign currency risks have the strongest explanatory and predictive power for crises, which match our approach and results. One of our key results is that NPL rates, credit stock dynamics, and foreign currency related exposures among NFIs are key warning signals. This is consistent with a wide range of studies, emphasizing the importance of credit quality and currency exposures in crisis prediction.

Consequently, we can provide a series of recommendations, as well as suggestions for further research. Firstly, we suggest incorporating the NFI heatmap into the National Bank of Romania’s (NBR) and National Committee for Macroprudential Oversight (NCMO) regular surveillance toolkit. Communicating the findings using the heatmap would give policymakers and analysts a clear visual of emerging vulnerabilities across categories. Moreover, the indicators and thresholds should be periodically recalculated, as insights will become more accurate over time. This is because now, our time frame is limited to January 2007 until June 2025, which only covers three financial crises. Therefore, the data should be updated regularly, as structural or political conditions change. Lastly, in terms of further research, we encourage experts to continue identifying relevant sub-sectors of the financial system, and analyze their corresponding indicators. A similar approach can be applied in order to assess the effects of climate change on financial stability, or the influence of other sectors such as insurance and investment funds. Similarly, building a heatmap for the banking sector or for the entire financial system could complement our NFI focus and help isolate cross-sector spillovers.

In conclusion, our heatmap framework has demonstrated the ability to flag periods of elevated risk in Romania, focusing on non-bank financial institutions. Information on past crises suggests it would have signaled vulnerabilities in a timely manner, making it a valuable complement to existing tools. By consistently tracking the chosen credit-risk and foreign currency-risk indicators, Romanian authorities can obtain early warning signals for pockets of risk in the NFI sector.

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6. Appendix

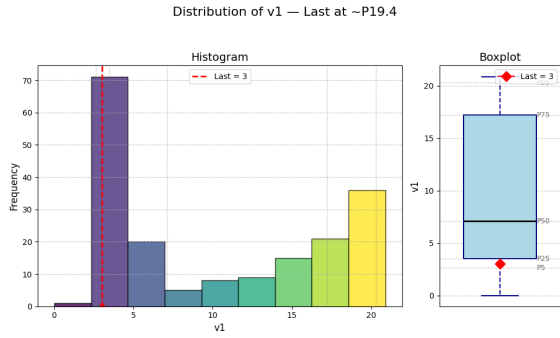


Figure 8: Histogram and Boxplot for v1 - Companies in Total

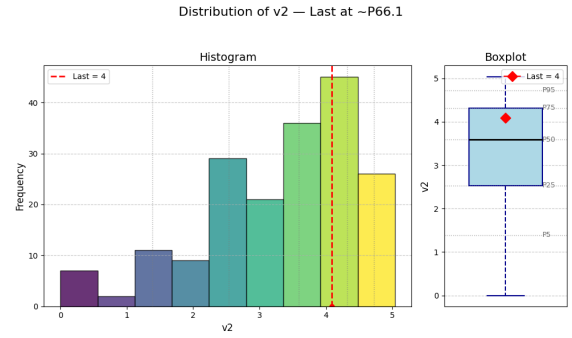


Figure 9: Histogram and Boxplot for v2 - Companies in Total

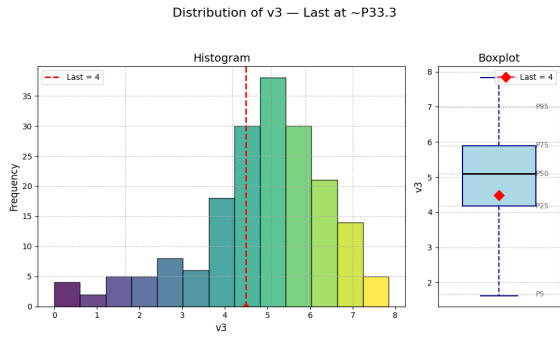


Figure 10: Histogram and Boxplot for v3 - Companies in Total

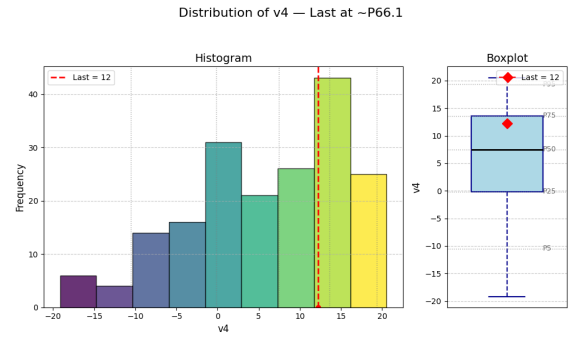


Figure 11: Histogram and Boxplot for v4 - Companies in Total

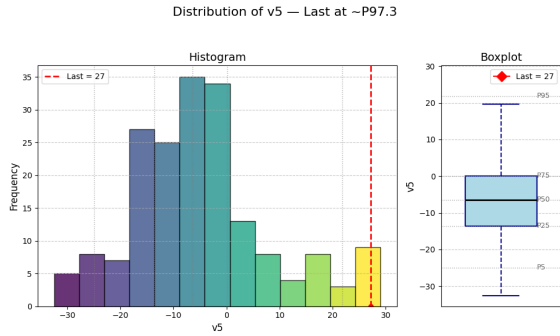


Figure 12: Histogram and Boxplot for v5 - Companies in Total

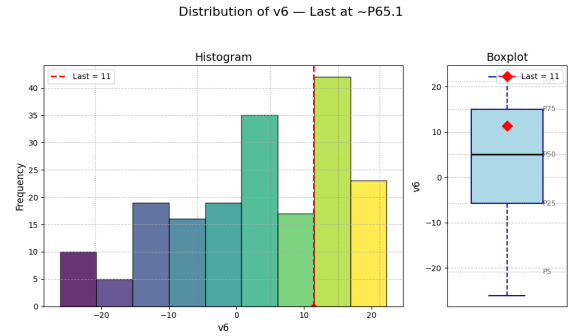


Figure 13: Histogram and Boxplot for v6 - Companies in Total

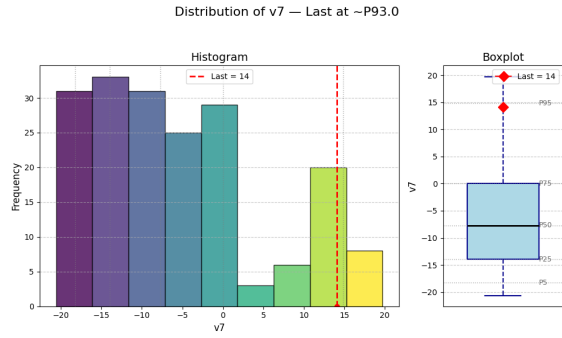


Figure 14: Histogram and Boxplot for v7 - Companies in Total

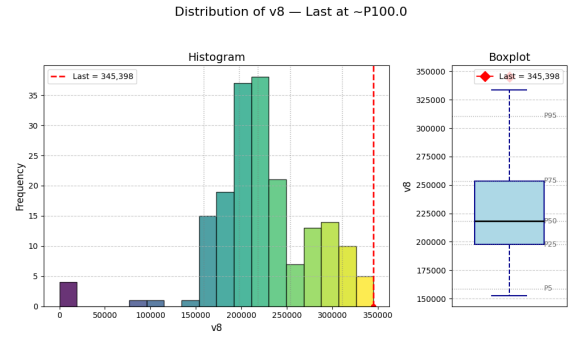


Figure 15: Histogram and Boxplot for v8 - Companies in Total

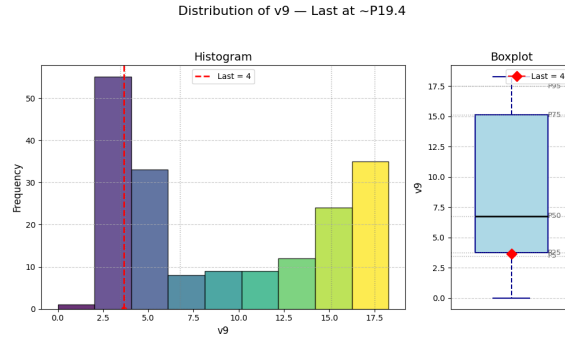


Figure 16: Histogram and Boxplot for v9 - Companies in Total

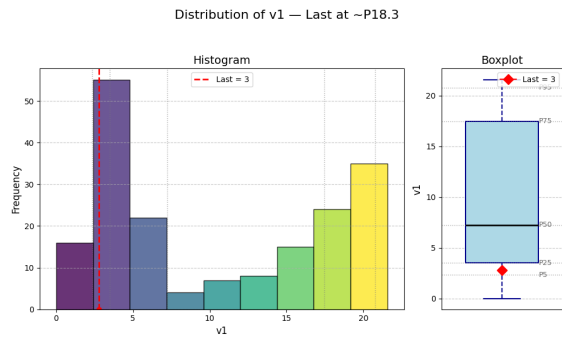


Figure 17: Histogram and Boxplot for v1 - Companies in Foreign Currency

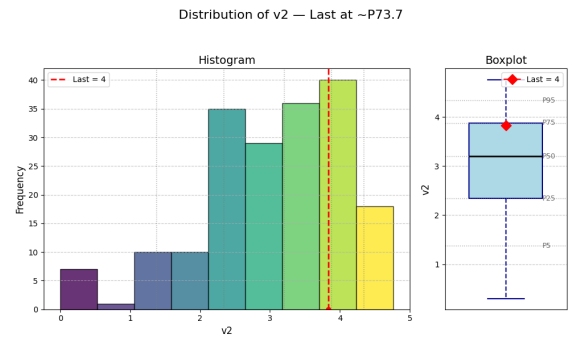


Figure 18: Histogram and Boxplot for v2 - Companies in Foreign Currency

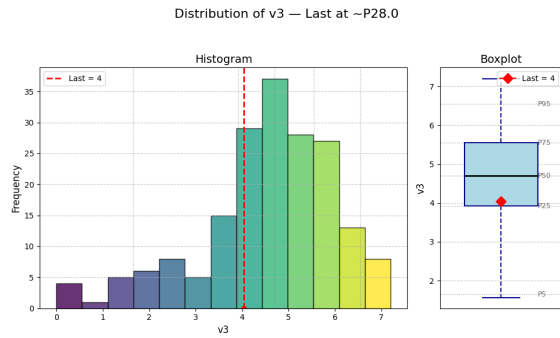


Figure 19: Histogram and Boxplot for v3 - Companies in Foreign Currency

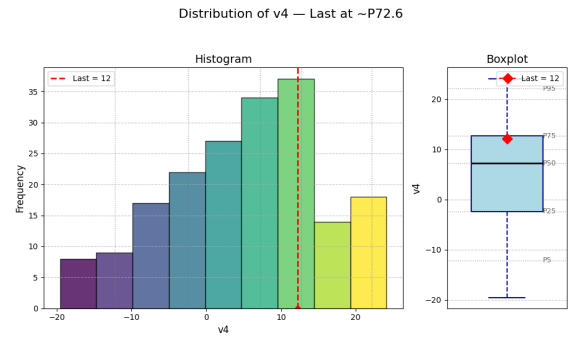


Figure 20: Histogram and Boxplot for v4 - Companies in Foreign Currency

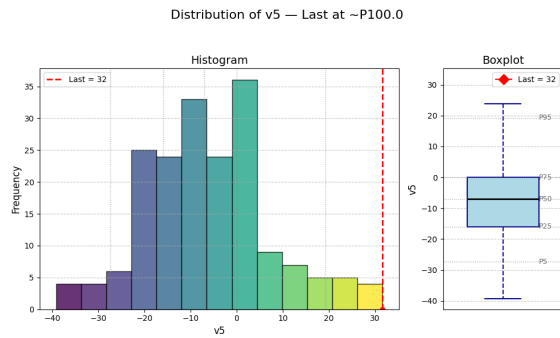


Figure 21: Histogram and Boxplot for v5 - Companies in Foreign Currency

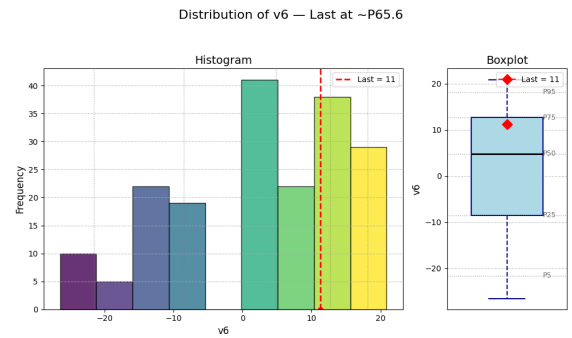


Figure 22: Histogram and Boxplot for v6 - Companies in Foreign Currency

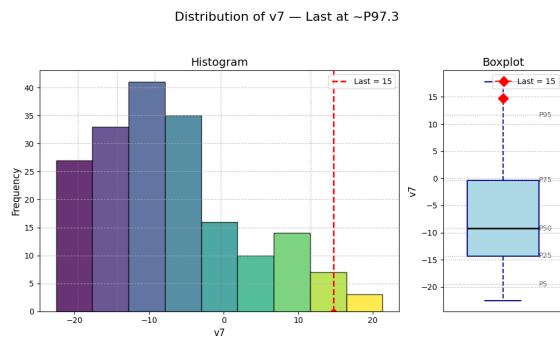


Figure 23: Histogram and Boxplot for v7 - Companies in Foreign Currency

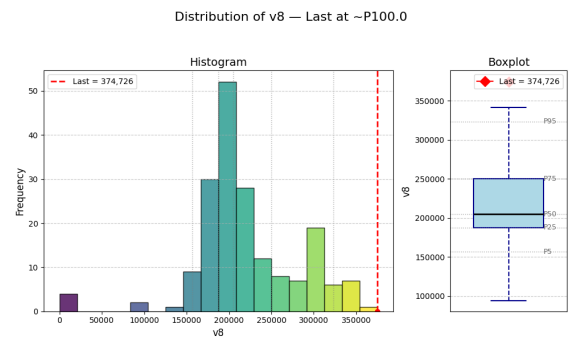


Figure 24: Histogram and Boxplot for v8 - Companies in Foreign Currency

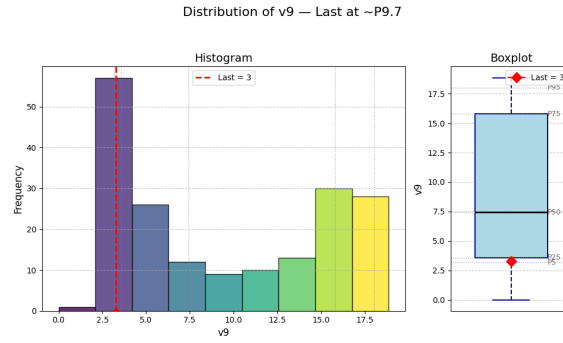


Figure 25: Histogram and Boxplot for v9 - Companies in Foreign Currency

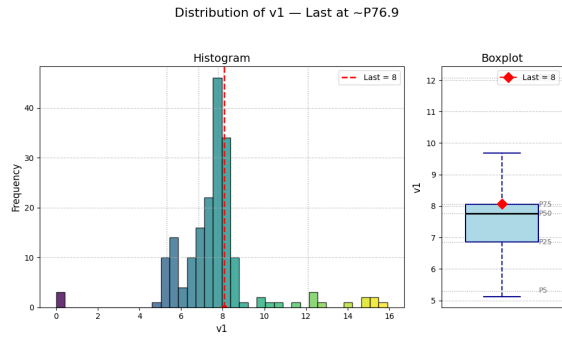


Figure 26: Histogram and Boxplot for v1 - Individuals in Total

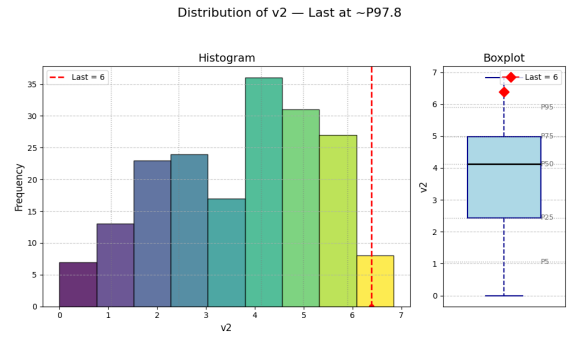


Figure 27: Histogram and Boxplot for v2 - Individuals in Total

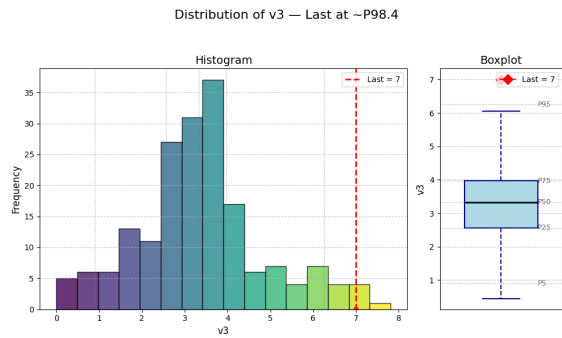


Figure 28: Histogram and Boxplot for v3 - Individuals in Total

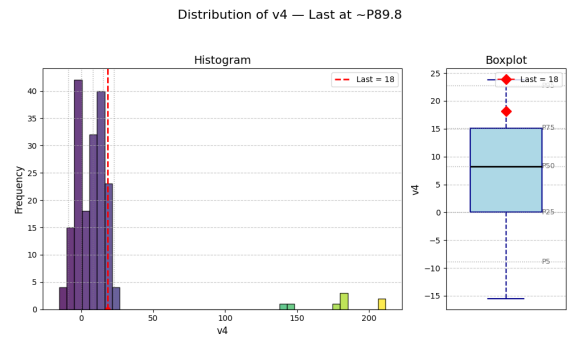


Figure 29: Histogram and Boxplot for v4 - Individuals in Total

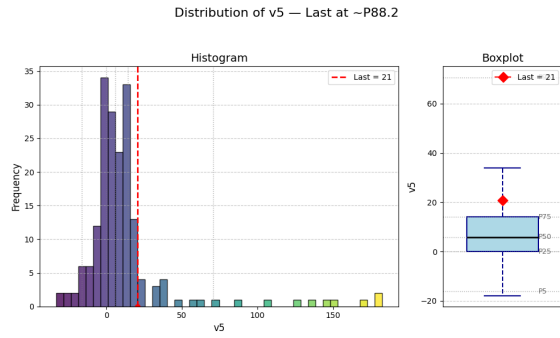


Figure 30: Histogram and Boxplot for v5 - Individuals in Total

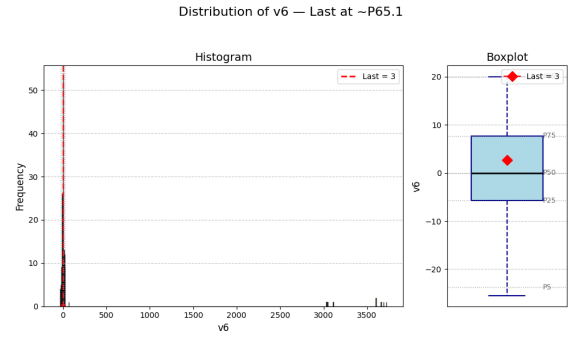


Figure 31: Histogram and Boxplot for v6 - Individuals in Total

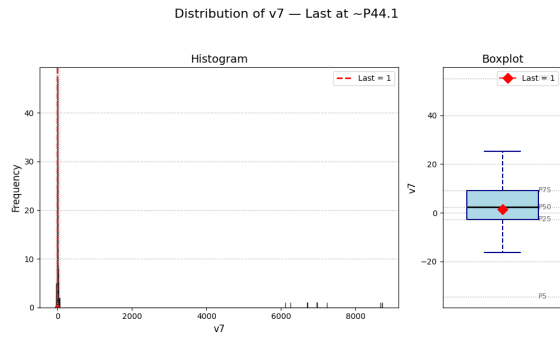


Figure 32: Histogram and Boxplot for v7 - Individuals in Total

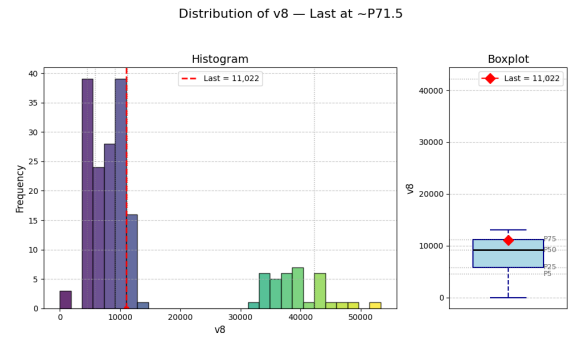


Figure 33: Histogram and Boxplot for v8 - Individuals in Total

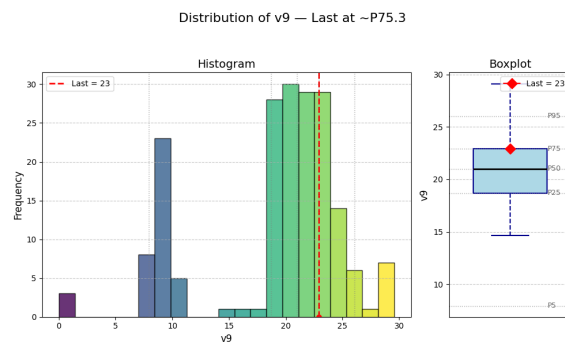


Figure 34: Histogram and Boxplot for v9 - Individuals in Total

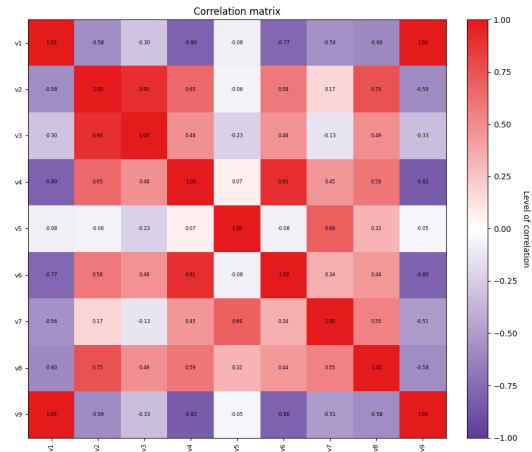


Figure 35: Correlation matrix for the 9 indicators - Companies in Total

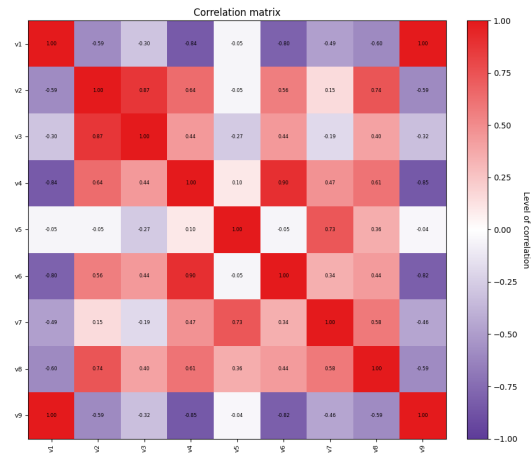


Figure 36: Correlation matrix for the 9 indicators - Companies in Foreign Currency

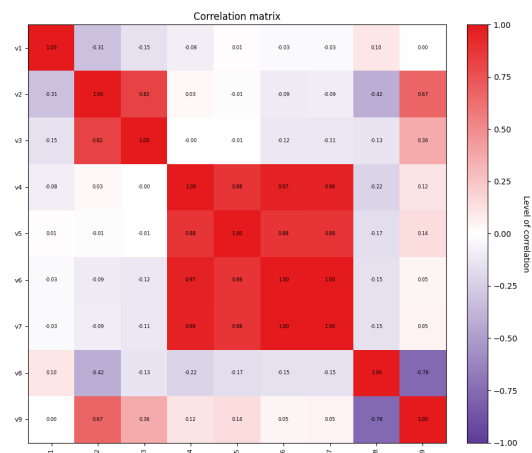


Figure 37: Correlation matrix for the 9 indicators - Individuals in Total

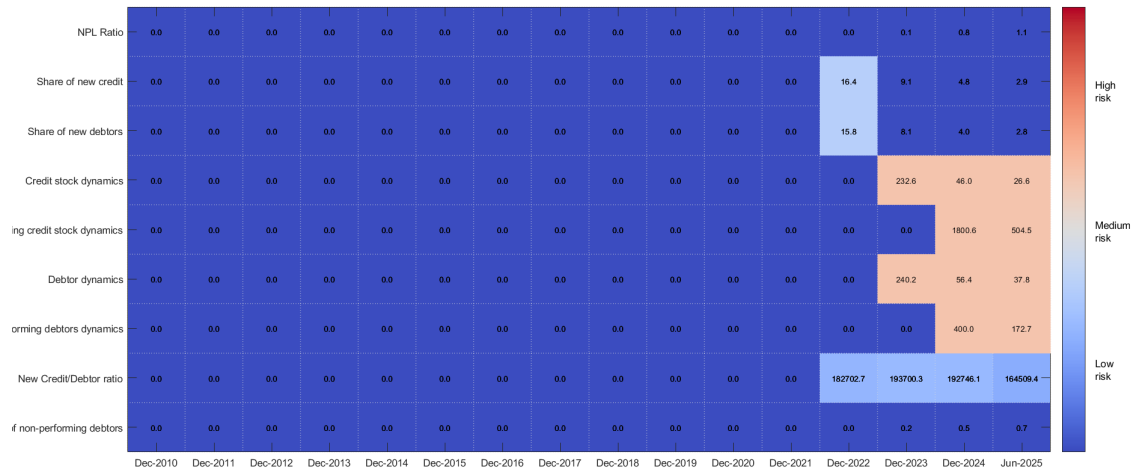


Figure 38: Additional Heatmap - 9 crediting risk indicators for Green Credit

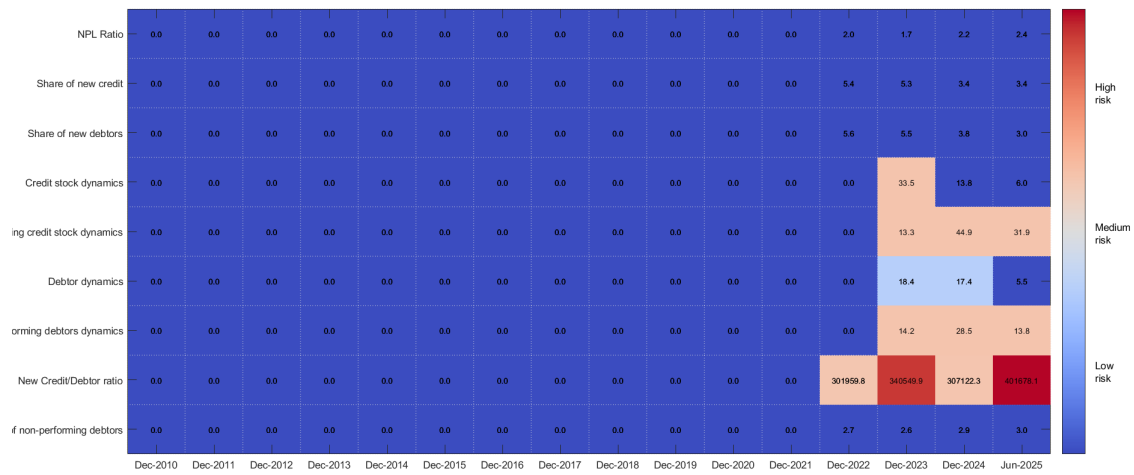


Figure 39: Additional Heatmap - 9 crediting risk indicators for Unstable Interest Rate

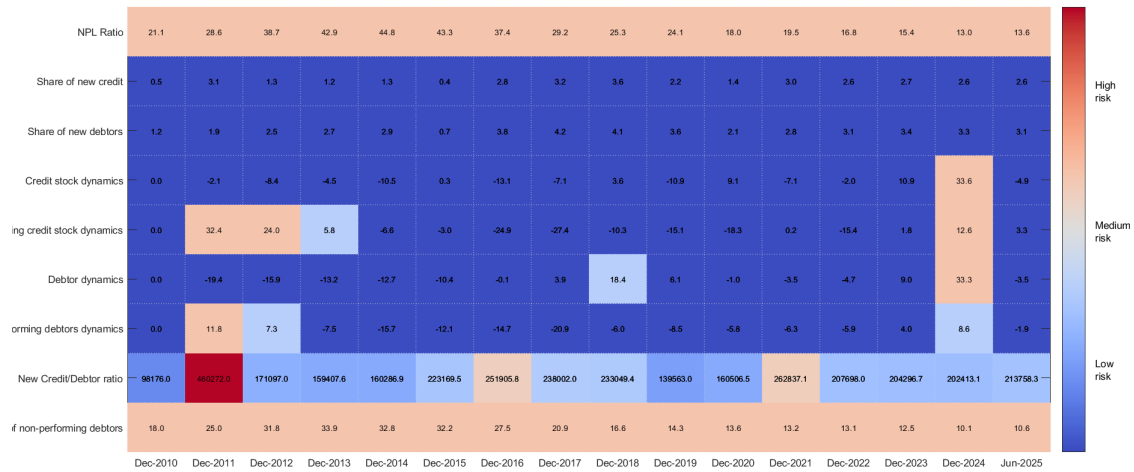


Figure 40: Additional Heatmap - 9 crediting risk indicators for Zombie Evaluated Companies

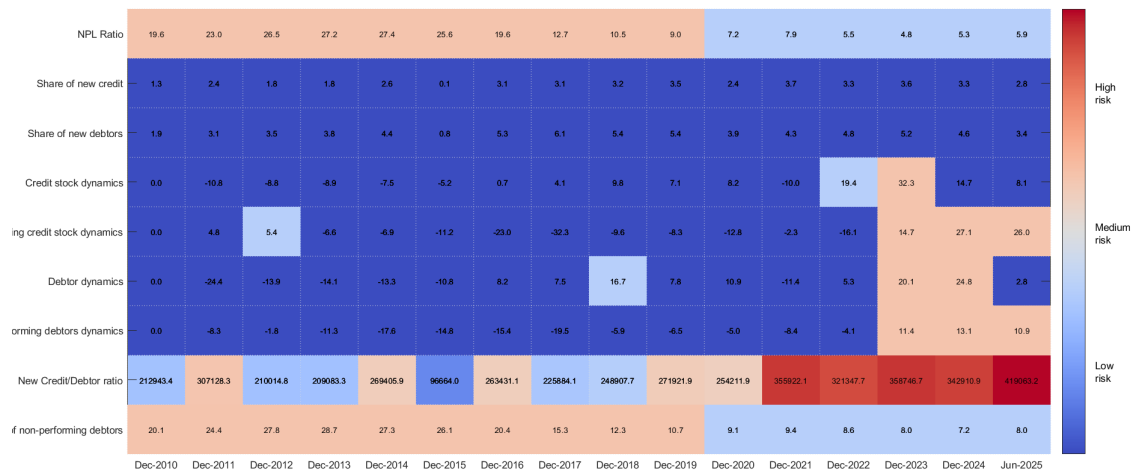


Figure 41: Additional Heatmap - 9 crediting risk indicators for Z-Score Evaluated Companies